

Flow Navigator

PROJECT MANAGER - seq_moore_over

Project Summary x seq_moore_over x
C:/Users/utkar/seq_moore_over/seq_moore_over.srscs/sources_1/new/seq_moore_over.v

```
1 timescale 1ns / 1ps
2 module seq_moore_over (din,clk,rst,state,out);
3 input din,clk,rst;
4 output reg [2:0] state;
5 output reg out;
6
7 parameter S0=3'b000,
8           S1=3'b001,
9           S2=3'b010,
10          S3=3'b011,
11          S4=3'b100,
12          S5=3'b101;
13
14 reg [26:0] counter;
15 reg clk2hz;
16 always @(posedge clk or posedge rst) begin
17     if (rst) begin
18         counter <= 0;
19         clk2hz <= 0;
20     end else begin
21         if (counter == 27'd99_999_999) begin
22             counter <= 0;
23             clk2hz <= ~clk2hz; // toggle every 1 sec -> 0.5 Hz clock (period = 2 sec)
24         end else begin
25             counter <= counter + 1;
26         end
27     end
28 end
29
30 always@(posedge clk2hz or posedge rst)
31 begin
32     if (rst)
```

Tcl Console Messages Log Reports Design Runs

44.70 Insert Verilog



Flow Navigator

PROJECT MANAGER

Settings

Add Sources

Language Templates

IP Catalog

IP INTEGRATOR

Create Block Design

Open Block Design

Generate Block Design

SIMULATION

Run Simulation

RTL ANALYSIS

Run Linter

Open Elaborated Design

SYNTHESIS

Run Synthesis

Open Synthesized Design

IMPLEMENTATION

Run Implementation

Open Implemented Design

PROJECT MANAGER - seq_moore_over

Project Summary

seq_moore_over.v

C:/Users/utkar/seq_moore_over/seq_moore_over.srcs/sources_1/new/seq_moore_over.v

```
25      counter <= counter + 1;
26      end
27      end
28      end
29
30      always@(posedge clk2hz or posedge rst)
31      begin
32      if (rst)
33      state<=S0;
34      else begin
35      case (state)
36      S0: state<=(din==1)?S1:S0;           //11011 moore overlapping sequence detection
37      S1: state<=(din==1)?S2:S0;
38      S2: state<=(din==0)?S3:S1;
39      S3: state<=(din==1)?S4:S0;
40      S4: state<=(din==1)?S5:S0;
41      S5: state<=(din==1)?S2:S3;
42      default: state<=S0;
43      endcase
44      end
45      end
46
47      always@(*)
48      begin
49      case (state)
50      S5: out=1'b1;
51      default: out=1'b0;
52      endcase
53      end
54      endmodule
55
```

Tcl Console

Messages

Log

Reports

Design Runs

11:28

Insert

Vivado

